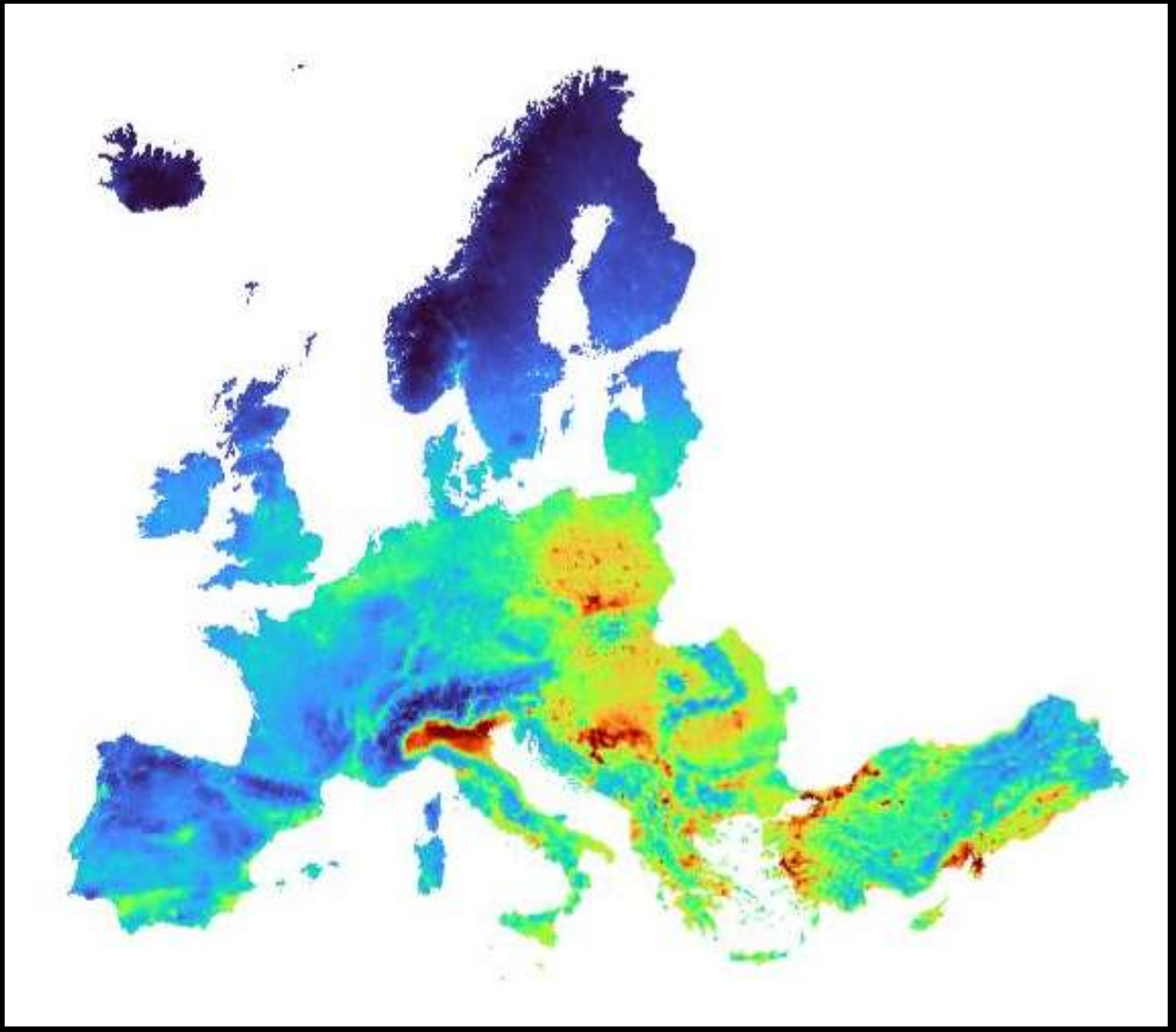
Uneven Air Pollution Exposure Across Europe (PM2.5, 2024)



Spatial distribution of PM2.5 concentrations reveals persistent regional inequality in air pollution exposure across Europe.

Data preprocessing steps:
Loaded PM2.5 GeoTIFF dataset into QGIS
Reprojected dataset to European coordinate system
Applied pseudocolor symbology (blue → red scale)
Visualized raster as continuous surface

Despite overall improvements in European air quality, spatial inequality remains strong, with Central and Eastern Europe experiencing significantly higher PM2.5 exposure than Northern Europe.

Spatial pattern analysis:
Clear geographic gradient in PM2.5 levels
Northern Europe shows consistently lower concentrations
Central and Eastern Europe show higher pollution zones
Urban and industrial regions appear as hotspots

DATA SOURCE
European Environment Agency
Dataset: PM2.5 European air quality (2024 interpolated)
Format: GeoTIFF raster (1 km grid)
Purpose: exposure estimation and air quality monitoring
Publication: June 2025